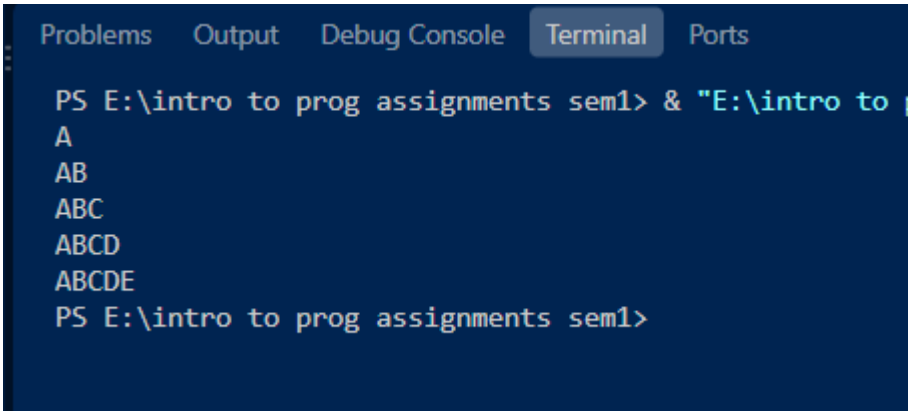
Q36

```c
#include<stdio.h>
int main()
{
    int end,temp,i,j;
    for(i=0; i<5; i++)
    {
        for(j=1; j <=i+1; j ++)
        {
        printf("%c", 64+j);
        }
    printf("\n");
    }
}
```



Q37

```c
#include<stdio.h>
int main()
{
    int i,j,m;
    for(i=0; i <5 ; i++)
    {
        for(j=1; j<=5-(i+1) ; j++)
        {
            printf(" ");
        }
        for(m=0; m<=i; m++)
        {
            printf("%d",m+1);
        }
        printf("\n");
    }
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS E:\intro to prog assignments sem1> & "E:\intro to prog assignments sem1\lab_5_que_37.exe"
    1
   12
  123
 1234
12345
PS E:\intro to prog assignments sem1>
```

Q38

Q39

```c
#include<stdio.h>
int main()
{
    int i,j,k,m;
    for(i=0;i<5;i++)
    {
        for(j=1;j<=i+1;j++)
        {
            printf("%c",65+i);
        }
        printf("\n");
    }
}
```

```
PS E:\intro to prog assignments sem1> & "E:\intro to prog assignments sem1\lab_5_que_39.exe"
A
BB
CCC
DDDD
EEEEE
PS E:\intro to prog assignments sem1>
```

Q40

```c
#include<stdio.h>
int main()
{
    int i,j;
    for(i=1; i<6; i++)
    {
        for(j=1; j<6; j++)
        {
```

```
            if(i==j)
            {
             printf("1");
            }
            else
            printf("0");
        }
        printf("\n");
    }
}
```

```
PS E:\intro to prog assignments sem1> & "E:\intro to prog assignments sem1\lav_5_que_37.exe"
10000
01000
00100
00010
00001
PS E:\intro to prog assignments sem1>
```

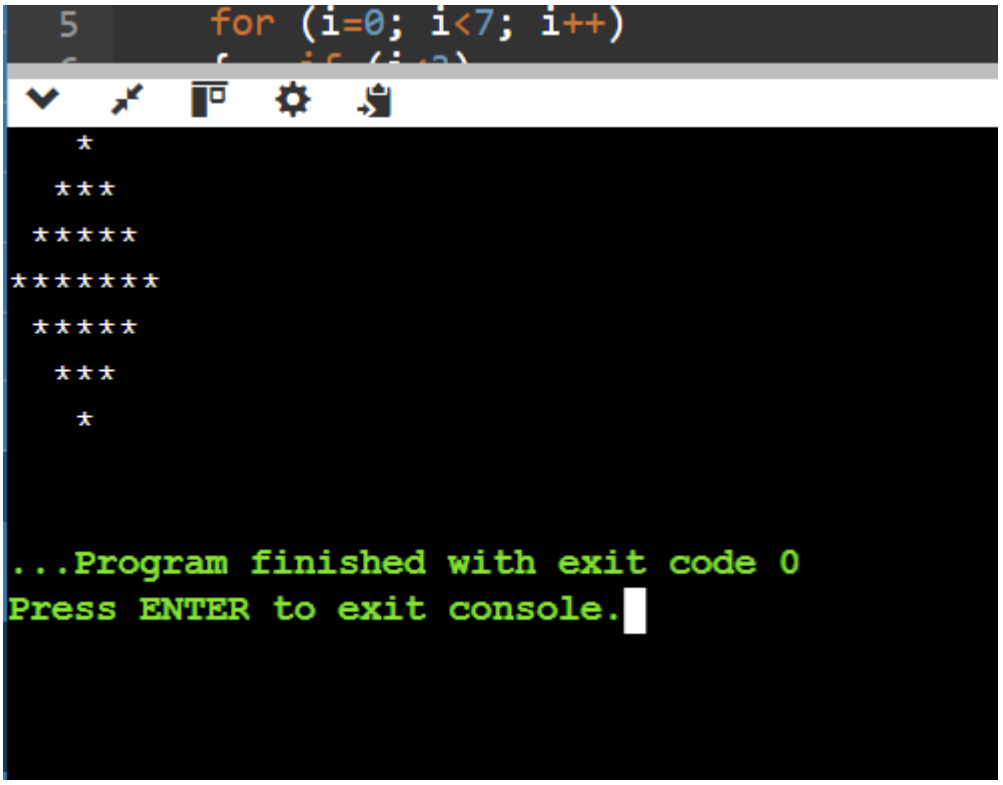
Q41

```
#include<stdio.h>
int main()
{
    int i,j;
    for(i=0;i<4;i++)
    {
        for(j=0;j<=4-(i+1) ; j++)
        {
            printf(" ");
        }
        for(j=1; j<=i+1; j++)
        {
            printf("%d", j);
        }
        for(j=i; j>=1; j--)
        {
            printf("%d", j);
        }
        printf("\n");
    }
}
```

```
PS E:\intro to prog assignments sem1> ./lab5qu
    1
   121
  12321
 1234321
PS E:\intro to prog assignments sem1>
```

Q42

```c
#include<stdio.h>
int main()
{
    int i,j;
    for (i=0; i<7; i++)
    {   if (i<3)
        {
            for(j=0; j<3-i; j++)
            {
                printf(" ");
            }
        }
        else if (i>3)
        {
            for(j=0; j<i-3; j++)
            {
                printf(" ");
            }
            for(j=0; j< 13-2*i; j++)
            {
            printf("*");
            }
        }
        if (i<4)
        for(j=0; j< 2*i+1; j++)
        {
            printf("*");
        }
        printf("\n");
    }
}
```



Q43

```c
#include<stdio.h>
int main()
{
    int arr[10],sum=0,i;
    for (i=0; i<10; i++)
    {
        printf("\ninput a number: ");
        scanf("%d", &arr[i]);
    }
    printf("\n numbers:");
    for (i=0; i<10; i++)
    {
        printf("%d\t", arr[i]);
    }
    for (i=0; i<10; i++)
    {
        sum = sum + arr[i];
    }
    printf("\nsum of given numbers: %d", sum);
}
```

```
input a number: 20

input a number: 10

 numbers:10     20      30      40      50      50      40      30      20      10      sum of given numbers: 300
PS E:\intro to prog assignments sem1>
```

Q44

```c
#include<stdio.h>
int main()
{
    int arr[10],odd=0,even=0,i;
    for (i=0;i<10;i++)
    {
        printf("\nenter a number: ");
        scanf("%d", &arr[i]);
    }
    printf("numbers:");
    for(i=0; i<10; i++)
    {
        printf("%d\t", arr[i]);
    }
    for (i=0; i<10; i++)
    {
        if ((arr[i])%2 == 0)
        {
            even++;
        }
        else if ((arr[i])%2 != 0)
        {
            odd++;
        }
    }
    printf("odd count: %d; even count: %d", odd, even);
}
```

```
enter a number: 43

enter a number: 2

enter a number: 1
numbers:120     12      11      144     11      5       66      43      2       1       odd count: 5; even count: 5
PS E:\intro to prog assignments sem1>
```

Q45

```c
#include<stdio.h>
int main()
{
    int arr[10],pos=0,neg=0,zeroes=0,i;
    for (i=0;i<10;i++)
    {
        printf("\nenter a number: ");
        scanf("%d", &arr[i]);
    }
    printf("numbers:");
    for(i=0; i<10; i++)
    {
        printf("%d\t", arr[i]);
    }
    for (i=0; i<10; i++)
    {
        if ((arr[i])>0)
        {
            pos++;
        }
        else if (arr[i]<0)
        {
            neg++;
        }
        else if(arr[i]==0)
        {
            zeroes++;
        }
    }
    printf("postive count: %d; negative count: %d; zeroes: %d", pos,
neg, zeroes);
}
```

```
enter a number: 44

enter a number: 32

enter a number: 99

enter a number: -11
numbers:120     121     11      -112    -112    123     44      32      99      -11     postive count: 7; negative count: 3; zeroes: 0
PS E:\intro to prog assignments sem1>
```

Q46

```c
#include<stdio.h>
int main()
```

```c
{
    int n;
    printf("enter amount of numbers:");
    scanf("%d", &n);
    int arr[n],max,smax,i;
    for (i=0;i<n;i++)
    {
        printf("\nenter a number: ");
        scanf("%d", &arr[i]);
    }
    printf("numbers:");
    for(i=0; i<n ;i++)
    {
        printf("%d\t", arr[i]);
    }
    if (arr[0] > arr[1])
    {
        max = arr[0];
        smax = arr[1];
    } else
    {
        max = arr[1];
        smax = arr[0];
    }
    for (i = 2; i < n; i++)
    {
        if (arr[i] > max)
        {
            smax = max;
            max = arr[i];
        }
        else if (arr[i] > smax && arr[i] != max)
        {
            smax = arr[i];
        }
    }
    printf("maximum: %d; second maximum: %d", max, smax);
}
```

```
enter a number: 12

enter a number: 19

enter a number: 13

enter a number: 4
numbers:12      19      13      4       maximum: 19; second maximum: 13
PS E:\intro to prog assignments sem1>
```

Q47

```c
#include<stdio.h>
int main()
{
    int first[10], secnd[10], third[10], i;
    printf("enter 10 numbers:");
    for(i=0; i<10; i++)
    {
        printf("\nenter a number:");
        scanf("%d", &first[i]);
    }
    printf("\nenter 10 numbers:");
    for(i=0; i<10; i++)
    {
        printf("\nenter a number:");
        scanf("%d", &secnd[i]);
    }
    printf("first entry:");
    for(i=0; i<10; i++)
    {
        printf("\t%d", first[i]);
    }
    printf("second entry:");
    for(i=0 ; i<10; i++)
    {
        printf("\t%d", secnd[i]);
    }
    for(i=0; i<10; i++)
    {
        third[i] = secnd[i] + first[i];
    }
    printf("\naddition: ");
    for (i =0; i<10; i++)
```

```
    {
        printf("%d\t", third[i]);
    }
}
```

```
enter a number:10
first entry:   10    9    8    7    6    5    4    3    2    1second entry: 1    2    3    4    5    6    7    8 9
10
addition: 11   11    11    11    11    11    11    11    11    11
PS E:\intro to prog assignments sem1>
```

Q48

```
#include<stdio.h>
int main()
{
    int or1[10], or2[10], temp, i;
    printf("enter first 10 numbers:");
    for(i=0; i<10; i++)
    {
        printf("\nenter a number:");
        scanf("%d", &or1[i]);
    }
    printf("enter next 10 numbers:");
    for(i=0; i<10; i++)
    {
        printf("\nenter a number:");
        scanf("%d", &or2[i]);
    }
    printf("before:");
    printf("\n first:");
    for (i=0; i<10; i++)
    {
        printf("\t%d", or1[i]);
    }
        printf("\n second:");
    for (i=0; i<10; i++)
    {
        printf("\t%d", or2[i]);
```

```
    }
    for (i=0; i<10; i++)
    {
        temp = or1[i];
        or1[i] = or2[i];
        or2[i] = temp;
    }
    printf("\nswapped:");
    printf("\n first:");
    for (i=0; i<10; i++)
    {
    printf("\t%d", or1[i]);
    }
    printf("\n second:");
    for (i=0; i<10; i++)
    {
        printf("\t%d", or2[i]);
    }
}
```

```
enter a number:20
before:
 first: 1       2       3       4       5       6       7       8       9       10
 second:        11      12      13      14      15      16      17      18      19      20
swapped:
 first: 11      12      13      14      15      16      17      18      19      20
 second:        1       2       3       4       5       6       7       8       9       10
PS E:\intro to prog assignments sem1>
```

Q49

```
#include<stdio.h>
int main()
{
    int i,j,k,n,temp;
    printf("enter the size of array:");
    scanf("%d", &n);
    int array[n];
    for(i=0;i<n;i++)
    {
        printf("enter an element:");
```

```
        scanf("%d",&array[i]);
    }
    printf("array:");
    for(i=0; i<n; i++)
    {
        printf("%d\t",array[i]);
    }
    printf("\n");
    for(i=0; i<n-1; i++)
    {
        for(j=i+1; j<n; j++)
        {
            if(array[i]>array[j])
            {
                temp = array[i];
                array[i] = array[j];
                array[j] = temp;
            }
        }
    }
    printf("\n selection sorted:");
    for(i=0;i<n;i++)
    {
        printf("\t%d", array[i]);
    }
}
```

```
enter an element:3
enter an element:7
enter an element:111
enter an element:18
array:14        5       632     -1      120     -20     11      13      3       7       111     18

 selection sorted:      -20     -1      3       5       7       11      13      14      18      111     120     632
PS E:\intro to prog assignments sem1>
```

Q50

```c
#include<stdio.h>
int main()
{
    int k,l,m,n,temp;
    printf("enter amount of numbers to be sorted:
");
    scanf("%d",&n);
    int ar[n];
    for(k=0;k<n;k++)
    {
      printf("\nenter a number:");
      scanf("%d",&ar[k]);
    }
    printf("numbers:");
    for (m=0; m<10; m++)
    {
        printf("\t %d",ar[m]);
    }
    for(l=0;l<n-1;l++)
    {
        for(m=l; m<n; m++)
        {
            if(ar[m]>ar[m+1])
            {
                temp = ar[m];
                ar[m] = ar[m+1];
```

```
                    ar[m+1] = temp;
                }
            }
        }
        printf("bubble sorted:");
        for (m=0; m<10; m++)
        {
            printf("\t%d",ar[m]);
        }
}
```

```
enter an element:56
array:100       82      -21     -12     1       9       56

 selection sorted:      -21     -12     1       9       56      82      100
PS E:\intro to prog assignments sem1>
```